

Date : 1 November 2022
Time : 2 hours

Marks available : 70
Full Marks : $66\frac{2}{3}$

*You may answer the questions in any order, provided your solutions are clearly labelled.
Full answers are expected. Please write neatly.
The use of electronic devices is forbidden; this includes calculators.*

1. For each of the following, either prove that the statement is true or give a reason why it is false. You can use the fact that $\sqrt{2}$ is an irrational number.
- (a) $\sqrt{\sqrt{2}-1}$ is an irrational number.
 - (b) Every bounded countable subset of $\mathbb{R}$ has a limit point.
 - (c) If a function f is continuous on an interval $[a, b]$, where $a < b$, then there is at least one point $x \in (a, b)$ such that $f(x)$ is a rational number. [7]
2. Let $(a_k)_{k=1}^{\infty}$ be the sequence defined by

$$a_1 = 3, \\ a_{k+1} = 2 + \sqrt{3a_k} \quad \text{for all } k \in \mathbb{N}.$$

- (a) Prove that $2 \leq a_k \leq a_{k+1} \leq 27$ for all $k \in \mathbb{N}$.
 - (b) Which theorem allows you to deduce that $(a_k)_{k=0}^{\infty}$ converges? Give the statement of this theorem and explain why it applies in this case.
 - (c) Find $\lim_{k \rightarrow \infty} a_k$. [5, 4, 6]
3. Let $(b_k)_{k=1}^{\infty}$ be a sequence with $\lim_{k \rightarrow \infty} b_k = 3$. Prove directly from the definition of a limit that
- $$\lim_{k \rightarrow \infty} \frac{2k + b_k}{k + 1} = 2. \quad [9]$$
4. Which of the following series are convergent? Justify your answers.

(a) $\sum_{k=1}^{\infty} k \left(\frac{2}{3}\right)^k$

(b) $\sum_{k=1}^{\infty} \frac{k + \ln(k)}{k^3 + k + 1}$

(c) $\sum_{k=1}^{\infty} 3^{1/k}$

[9]

5. For each of the following, determine whether the given series is absolutely convergent, conditionally convergent or divergent:

(a) $\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln(k)}$

(b) $\sum_{k=2}^{\infty} \left(\frac{-1}{\ln(k)} \right)^k$

In each case, give full reasons for your answer.

[9]

6. For each $k \in \mathbb{N}$, let $h_k(x) = e^{-kx}$ for all $x \in \mathbb{R}$.

(a) What is the pointwise limit h of the sequence $(h_k)_{k=1}^{\infty}$ on the interval $[0, \infty)$?

(b) Does $(h_k)_{k=1}^{\infty}$ converge to h uniformly on $[0, 1]$?

[3, 3]

7. Compute the radius of convergence of each of the following power series:

(a) $\sum_{k=1}^{\infty} \frac{x^k}{k^4 4^k}$

(b) $\sum_{k=0}^{\infty} \frac{2^k}{3^k + (-1)^k 4^k} (x-5)^k$

[4]

8. Read the following theorem and the outline of its proof:

Theorem. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is a continuous function, then for each non-empty compact set $K \subseteq \mathbb{R}$, the set $f^{-1}(K) = \{x \in \mathbb{R} : f(x) \in K\}$ is compact.

Proof. Let $(y_n)_{n=1}^{\infty}$ be any sequence in $f^{-1}(K)$. For each $n \in \mathbb{N}$, choose a point $x_n \in K$ with $f(x_n) = y_n$. Then $(x_n)_{n=1}^{\infty}$ has a subsequence $(x_{n_k})_{k=1}^{\infty}$ that converges to a point $x \in K$. Then $\lim_{k \rightarrow \infty} y_{n_k} = f(x)$, so $f^{-1}(K)$ is compact. $\square$

Now answer the following questions:

- (a) Define what is meant by a non-empty set $K \subseteq \mathbb{R}$ being *compact*.
- (b) Explain why it is possible to choose an $x_n \in K$ with $f(x_n) = y_n$.
- (c) Towards the end of the proof, it is claimed that $\lim_{k \rightarrow \infty} y_{n_k} = f(x)$, but no explanation is given. While it is possible to establish this result by using the Sequential Criterion for Continuity, we would like you to prove it directly using the (ε, δ) definition of a continuous function.
- (d) Explain why it follows that $f^{-1}(K)$ is compact.

[1, 1, 6, 3]